Solution Manual

Analysis and Design of Elementary MOS Amplifier Stages

- First Edition -
Chapter 2

Boris Murmann
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$restart : with(Units[Standard]) : with(plots) : interface(imaginaryunit = j) : \textbf{read}$ "Parameters.mpl" :

2.1

$$200[\![uA]\!] = \frac{1}{2}\mu C_{ox}\cdot 10\cdot\left(1.5[\![V]\!] - V_T\right)^2 :$$

$$50[\![uA]\!] = \frac{1}{2}\mu C_{ox}\cdot 10\cdot\left(1.0[\![V]\!] - V_T\right)^2 :$$

Dividing these two expressions and taking the square root gives

$$2 = \frac{1.5 - V_T}{1.0 - V_T} :$$

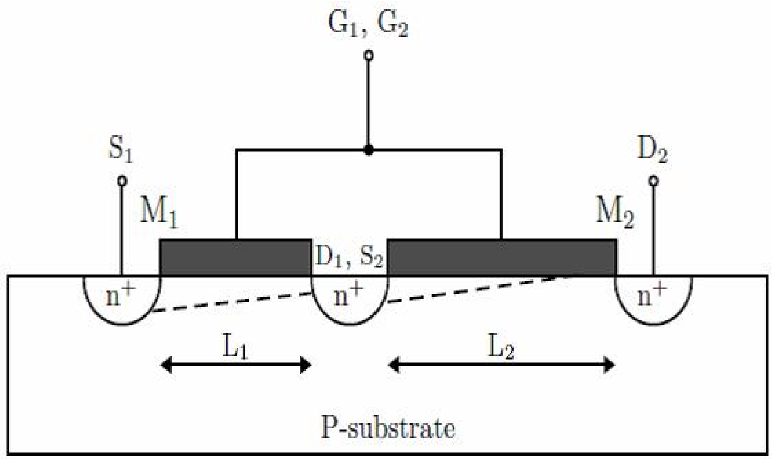
$$solve\left(2 = \frac{1.5 - V_T}{1.0 - V_T}, V_T\right) = 0.500 \text{ [V]}$$

We can now easily compute uCox using either one of the current equations.

$$\mu C_{ox} := \frac{2\cdot 200[\![uA]\!]}{10\cdot(1.5[\![V]\!] - 0.5[\![V]\!])^2} = 0.000\left[\!\!\left[\frac{A^3\, s^6}{m^4\, kg^2}\right]\!\!\right] \xrightarrow{\text{replace units}} 40.000\left[\!\!\left[\frac{uA}{V^2}\right]\!\!\right]$$

2.2

The preferred way to solve this problem is by inspecting the cross-sectional view of the given structure (see below). As depicted in this figure, the overall structure looks like a transistor with an effective channel length of L1 + L2. This happens because all the electrons that enter the channel at S1 have to go through both channels. Note that the diffusion area in between (D1 or S2) does not affect carriers movement since it is assumed to be highly-doped. Furthermore, the transistor on the left cannot be in saturation because once channel 1 pinches off, the other transistor will turn off. Hence, the left transistor can only operate in triode region. Putting these arguments together one can conclude that this structure is just a transistor whose channel is the sum of two smaller channels of length L1 and L2.



A more algebra-intense method is to study the given structure as a three-terminal device and consider its behavior under different bias conditions using the respective I-V laws:

1.VGS < Vt: both devices are off.

2.VGS > Vt and VDS < VGS - Vt. In this case, it can be shown that both transistors must operate in the triode region. We can use use algebra to combine the two I-V laws and show that the device behaves like a single transistor with L=L1+L2.

3. VGS > Vt and VDS > VGS - Vt. In this case, we can show algebraically that one transistor must be in triode and the other in saturation. Again, we can then use algebra to combine the two I-V laws and show that the device behaves like a single transistor with L=L1+L2.

An important take-home from this problem is that it is impossible for both transistors to be saturated. This is obvious form the above drawing, but somewhat painful to show with algebra (give it a try, if you like).

2.3

In point C of Figure 2-7(b), the output voltage is equal to VIN-VT.

$$V_{OUT} = V_{DD} - R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L} \cdot \left(V_{IN} - V_T\right)^2 = V_{IN} - V_T :$$

$$V_{DD} - a \cdot x^2 = x :$$

$$x^2 + \frac{x}{a} - \frac{V_{DD}}{a} = 0 :$$

$$x_{1,2} = -\frac{1}{2a} \pm \sqrt{\frac{1}{4a^2} + \frac{V_{DD}}{a}} :$$

The (-) solution can be discarded, since it corresponds to negative VIN-VT (transistor is off).

$$x = -\frac{1}{2a} + \sqrt{\frac{1}{4a^2} + \frac{V_{DD}}{a}} :$$

$$V_{IN} - V_T = -\frac{1}{2 R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L}} + \sqrt{\frac{1}{4\left(R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L}\right)^2} + \frac{V_{DD}}{R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L}}} :$$

$$V_{IN} = V_T + \frac{1}{2 R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L}} \left(\sqrt{1 + 4 \cdot V_{DD} \cdot R_D \cdot \frac{1}{2} \cdot \mu C_{oxn} \cdot \frac{W}{L}} - 1\right) :$$

From this expression, we see that VIN (in point C) approaches VT as RD increases.

2.4

$V_{DD} := 5 [\![V]\!] : R_{\mathrm{D}} := 10 [\![k\Omega]\!] : W := 10 [\![um]\!] : L := 1 [\![um]\!] : V_{IN} := 1.5 [\![V]\!] :$

We can solve this problem using the expression derived in problem 2.3:

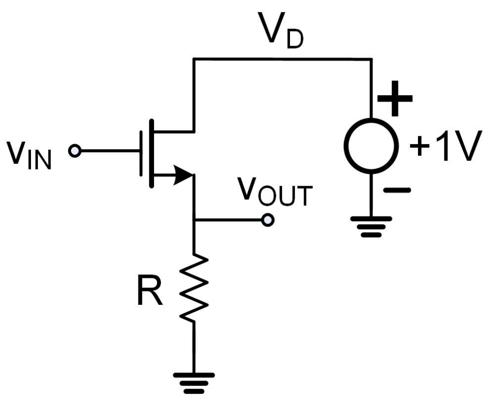
$$v_{IN} := V_{T0n} + \frac{1}{2\, R_{\mathrm{D}} \cdot \frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L}} \left(\sqrt{1 + 4. \cdot V_{DD} \cdot R_{\mathrm{D}} \cdot \frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L}} - 1 \right) = 1.728 \ [\![V]\!]$$

$$v_{in} := v_{IN} - V_{IN} = 0.228 \ [\![V]\!]$$

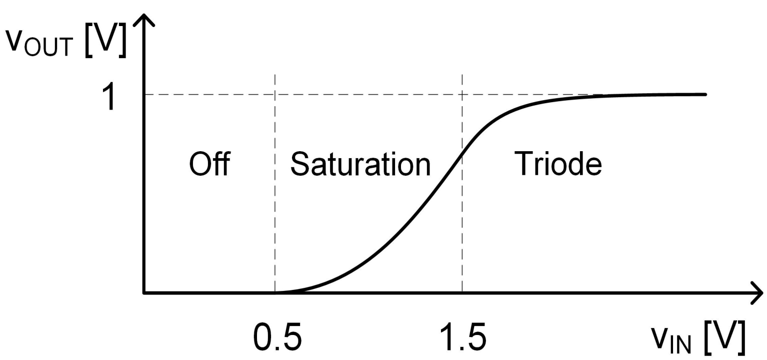
Therefore, the input voltage can be increased by 228 mV (from the operating point) before the MOSFET enters the triode region.

2.5

For small values of v_{IN} there is no current through the device and therefore v_{OUT} is zero. Since the drain terminal in an NMOS is at a higher potential than the source terminal, for v_{OUT} < 1V the drain and source swap places and we can redraw the circuit as shown below. For v_{IN} < VTn = 0.5 V the device is off. Increasing v_{IN} beyond 0.5 V will put the transistor into the saturation region. In this region the current and therefore v_{OUT} have a parabolic shape determined by the square law formula.



The transistor enters the triode region when VGS-VTn = VDS, i.e. v_{IN}-v_{OUT}-V_{Tn} = V_D-v_{OUT}, which leads to v_{IN} = V_D+V_{Tn}=1.5V. From then on, the output voltage must approach 1V, as shown below (the on-resistance of the transistor decreases monotonically as v_{IN} continues to rise).



2.6

This problem is obsolete since it is already solved in Chapter 2 (page 31), see also errata.

2.7

Note: The problem statement has a typo. The problem below is worked for K = uCoxW/L = 100 uA/V^2.5 (see also errata).

$$K := 100 \left[\!\left[\frac{uA}{V^{2.5}} \right]\!\right] : I_{\mathrm{D}} := 1 \left[\!\left[uA \right]\!\right] :$$

$$g_m = \frac{d}{dV_{GS}} \left(\frac{1}{2} K \cdot \left(V_{GS} - V_T \right)^{2.5} \right) = \frac{2.5}{2} K \cdot \left(V_{GS} - V_T \right)^{1.5} :$$

$$V_{GS} - V_T = \left(\frac{I_{\mathrm{D}}}{\frac{1}{2} K} \right)^{\frac{1}{2.5}} :$$

$$g_m := \frac{5}{4} K \cdot \left(\frac{I_{\mathrm{D}}}{\frac{1}{2} K} \right)^{\frac{3}{5}} = 0.000 \left[\!\left[\frac{A^{2.000}\, s^{3.000}}{m^{2.000}\, kg^{1.000}} \right]\!\right] \xrightarrow{\text{replace units}} 11.954 \left[\!\left[uS \right]\!\right]$$

2.8

$$\frac{i_d}{I_D} = \frac{g_m v_{gs}}{I_D} = \frac{\dfrac{2 \cdot I_D}{V_{OV}} v_{gs}}{I_D} = 2 \cdot \frac{v_{gs}}{V_{OV}} :$$

2.9

$V_{IN} := 1.5[\![V]\!] : W := 20[\![um]\!] : L := 1[\![um]\!] : R_{\mathrm{D}} := 5[\![k\Omega]\!] : V_{DD} : 5[\![V]\!] :$

We need to check if the transistor is in saturation.

$$V_{OV} := V_{IN} - V_{T0n} =$$

$$I_{\mathrm{D}} := \frac{1}{2}\mu n C_{ox} \cdot \frac{W}{L} \cdot V_{OV}^{2} = 0.001\ [\![A]\!] \xrightarrow{\text{replace units}} 500.000\ [\![uA]\!]$$

$$V_{DS} := V_{DD} - R_{\mathrm{D}} \cdot I_{\mathrm{D}} = 2.500\ [\![V]\!]$$

The MOSFET is in saturation since VDS > VOV.

$$g_m := 2 \cdot \frac{I_{\mathrm{D}}}{V_{OV}} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 1.000\ [\![mS]\!]$$

$$A_v := -g_m \cdot R_{\mathrm{D}} = -5.000$$

2.10

$W := 20[\![um]\!] : L := 1[\![um]\!] : R_D := 5[\![k\Omega]\!] : V_{DD} : 5[\![V]\!] : V_{OUT} := 2.5[\![V]\!] :$

$V_{OUT} = -I_D \cdot R_D = \frac{1}{2}\,\mu p C_{ox} \cdot \frac{W}{L} \cdot V_{OV}^2 \cdot R_D :$

$$V_{OV} := \sqrt{\frac{2 \cdot \frac{V_{OUT}}{R_D}}{\mu p C_{ox} \cdot \frac{W}{L}}} = 1.414\ [\![V]\!]$$

$V_{OV} = V_{SG} + V_{Tp} = V_{DD} - V_{IN} + V_{Tp} :$
$V_{IN} := V_{DD} - V_{OV} + V_{T0p} = 3.086\ [\![V]\!]$

Check: VSD = VDD-VOUT = 2.5 V > VOV. The device operates in saturation, as assumed.

$$g_m := \frac{2\frac{V_{OUT}}{R_D}}{V_{OV}} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 0.707\ [\![mS]\!]$$
$A_v := -g_m \cdot R_D = -3.536$

2.11

$v_{OUT} := 'v_{OUT}' : V_{DD} := 'V_{DD}' : v_{IN} := 'v_{IN}' : V_{DS} := 'V_{DS}' : R_D := 'R_D' : V_{OUT} := 'V_{OUT}' :$

$$v_{OUT} = V_{DD} - R_D \cdot \mu C_{oxn} \cdot \frac{W}{L} \cdot \left(v_{IN} - V_{Tn} - \frac{1}{2} \cdot V_{DS} \right) \cdot V_{DS} :$$

$$v_{OUT} = V_{DD} - C \cdot \left(v_{IN} - V_{Tn} - \frac{1}{2} \cdot v_{OUT} \right) \cdot v_{OUT} :$$

We need to find dv_{OUT}/dv_{IN}. The shortest way to find this derivative is via implicit differentiation.

$$\frac{dv_{OUT}}{dv_{IN}} = -C \cdot \frac{d}{dv_{IN}} \left(v_{IN} \cdot v_{OUT} - V_{Tn} \cdot v_{OUT} - \frac{1}{2} \cdot v_{OUT}^2 \right) :$$

$$\frac{dv_{OUT}}{dv_{IN}} = -C \cdot \left(v_{OUT} + v_{IN} \cdot \frac{dv_{OUT}}{dv_{IN}} - V_{Tn} \cdot \frac{dv_{OUT}}{dv_{IN}} - \frac{1}{2} \cdot 2 \cdot \frac{dv_{OUT}}{dv_{IN}} v_{OUT} \right) :$$

$$\frac{dv_{OUT}}{dv_{IN}} = -C \cdot \left(v_{OUT} + \left(v_{IN} - V_{tn} \right) \cdot \frac{dv_{OUT}}{dv_{IN}} - \frac{dv_{OUT}}{dv_{IN}} v_{OUT} \right) :$$

$$\frac{dv_{OUT}}{dv_{IN}} \left(1 + C \cdot \left(v_{IN} - V_{tn} \right) - C \cdot v_{OUT} \right) = -C \cdot v_{OUT} :$$

$$\frac{dv_{OUT}}{dv_{IN}} = -\frac{C \cdot v_{OUT}}{\left(1 + C \cdot \left(v_{IN} - V_{tn} \right) - C \cdot v_{OUT} \right)} :$$

Quick check using Maple:

$$\mathit{implicitdiff}\left(v_{OUT} = V_{DD} - C \cdot \left(v_{IN} - V_{Tn} - \frac{1}{2} \cdot v_{OUT} \right) \cdot v_{OUT}, v_{OUT}, v_{IN} \right)$$

$$\frac{C\, v_{OUT}}{-1 + C\, v_{OUT} - C\, v_{IN} + C\, V_{Tn}} \qquad \textbf{(1)}$$

To get to the final answer, we must now evaluate the derivative at the operating point. Everything except VOUT is known. The shortest way to get to the answer is to plug in numbers (instead of symbolic analysis) and solve the resulting quadratic equation.

$$V_{OUT} = V_{DD} - R_D \cdot \mu C_{oxn} \cdot \frac{W}{L} \cdot \left(v_{IN} - V_{Tn} - \frac{1}{2} \cdot V_{OUT} \right) \cdot V_{OUT} :$$

$$V_{OUT} = 5 - 10 \cdot \left(1 - \frac{1}{2} \cdot V_{OUT} \right) \cdot V_{OUT} :$$

$$V_{OUT} = 5 - 10\, V_{OUT} + 5\, V_{OUT}^2 :$$

$$0 = 1 - \frac{11}{5}\, V_{OUT} + V_{OUT}^2 :$$

$$V_{OUT1} := \frac{11}{10} [\![V]\!] + \sqrt{\left(\frac{11}{10} \right)^2 - 1.}\ [\![V]\!] = 1.558\ [\![V]\!]$$

$$V_{OUT2} := \frac{11}{10} [\![V]\!] - \sqrt{\left(\frac{11}{10} \right)^2 - 1.}\ [\![V]\!] = 0.642\ [\![V]\!]$$

The first answer can be discarded, since the transistor would be in saturation: VDS=1.558V > VOV=1V.

$V_{IN} := 1.5[\![V]\!] : R_{\mathrm{D}} := 10[\![k\Omega]\!] : W := 20[\![um]\!] : L := 1[\![um]\!] :$

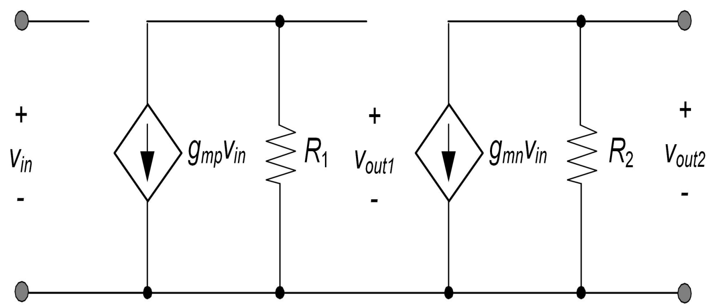
Finally we find:

$$A_v := -\frac{V_{OUT2}}{\left(\dfrac{1}{R_{\mathrm{D}}\cdot \mu nC_{ox}\cdot \dfrac{W}{L}} + \left(V_{IN} - V_{T0n}\right) - V_{OUT2}\right)} = -1.400$$

Note that this number is considerably smaller that the voltage gain obtained in problem 2.9, when the transistor operates in saturation.

2.12

(a) The small signal model can be constructed as follows.



(b) Av1=vout1/vin follows from:

$$A_{v1} = -g_{mp} \cdot R_1 = -\frac{2 \cdot I_D \cdot R_1}{V_{OV}} = -\frac{2 \cdot \frac{V_{DD}}{2}}{\frac{V_{DD}}{2} - V_{Tn}} = \frac{V_{DD}}{\frac{V_{DD}}{2} - V_{Tn}} \quad :$$

$$V_{DD} := 2.5 [\![V]\!] \ :$$

$$A_{v1} := -\frac{V_{DD}}{\frac{V_{DD}}{2} - V_{T0n}} = -3.333$$

The voltage gain vout2/vi is simply

$$A_{v2} := A_{v1}^2 = 11.111$$

2.13

$V_{IN} := 1.5[\![V]\!] : W := 40[\![um]\!] : L := 2[\![um]\!] : R_D := 5[\![k\Omega]\!] : V_{DD} := 5[\![V]\!] :$

We calculate the bias point without considering channel-length modulation. This gives the same numbers for ID, VDS and gm as in problem 2.9.

$$I_D := \frac{1}{2}\mu n C_{ox}\cdot\frac{W}{L}\cdot\left(V_{IN}-V_{T0n}\right)^2 = 0.001\ [\![A]\!] \xrightarrow{\text{replace units}} 500.000\ [\![uA]\!]$$

$$V_{DS} := V_{DD} - R_D\cdot I_D = 2.500\ [\![V]\!]$$

$$g_m := 2\cdot\frac{I_D}{V_{IN}-V_{T0n}} = 0.001\ [\![S]\!] \xrightarrow{\text{replace units}} 1.000\ [\![mS]\!]$$

When calculating the voltage gain, we must now consider the MOSFET's output resistance:

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.050\ \left[\!\left[\frac{1}{V}\right]\!\right]$$

$$r_o := \frac{1+\lambda_n\cdot V_{DS}}{\lambda_n\cdot I_D} = 45000.000\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 45.000\ [\![kOmega]\!]$$

$$A_v := -g_m\cdot\left(\frac{1}{R_D}+\frac{1}{r_o}\right)^{-1} = -4.500$$

Compared to problem 2.9, the voltage gain is reduced by about 10%.

2.14

$V_{OUT} := 2.5[\![V]\!] : W := 12[\![um]\!] : L := 2[\![um]\!] : R_D := 10[\![k\Omega]\!] : V_{DD} := 5[\![V]\!] :$

(a) We begin by computing the drain current at the bias point and the calculate VIN via VOV

$$I_D := -\frac{V_{OUT}}{R_D} = -0.000\ [\![A]\!] \xrightarrow{\text{replace units}} -250.000\ [\![uA]\!]$$

$$V_{OV} := \sqrt{\frac{-I_D}{1.\ \mu pC_{ox} \cdot \frac{W}{L}}} = 1.000\ [\![V]\!]$$

$$V_{OV} = V_{SG} + V_{Tp} = V_{DD} - V_{IN} + V_{Tp} :$$

$$V_{IN} := V_{DD} - V_{OV} + V_{T0p} = 3.209\ [\![V]\!]$$

(b) The voltage gain in presence of channel-length modulation is found as follows:

$$g_m := \frac{2 \cdot (-I_D)}{V_{OV}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 387.298\ [\![uS]\!]$$

$$V_{SD} := V_{DD} - V_{OUT} = 2.500\ [\![V]\!]$$

$$\lambda_p := \frac{\lambda_{Lp}}{L} = 0.050\ \left[\!\!\left[\frac{1}{V}\right]\!\!\right]$$

$$r_o := \frac{1 + \lambda_p \cdot V_{SD}}{\lambda_p \cdot (-I_D)} = 90000.000\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 90.000\ [\![kOmega]\!]$$

$$A_v := -g_m \cdot \left(\frac{1}{R_D} + \frac{1}{r_o}\right)^{-1} = -3.486$$

(c) We perform the same calculations as in parts (a) and and (b) with the new value for RD.

$R_D := 100[\![k\Omega]\!] :$

$$I_D := -\frac{V_{OUT}}{R_D} = -0.000\ [\![A]\!] \xrightarrow{\text{replace units}} -25.000\ [\![uA]\!]$$

$$V_{OV} := \sqrt{\frac{-I_D}{1.\ \mu pC_{ox} \cdot \frac{W}{L}}} = 0.408\ [\![V]\!]$$

$$V_{IN} := V_{DD} - V_{OV} + V_{T0p} = 4.092\ [\![V]\!]$$

$$g_m := \frac{2 \cdot (-I_D)}{V_{OV}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 122.474\ [\![uS]\!]$$

$$r_o := \frac{1 + \lambda_p \cdot V_{SD}}{\lambda_p \cdot (-I_D)} = 9.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 900.000\ [\![kOmega]\!]$$

$$A_v := -g_m \cdot \left(\frac{1}{R_D} + \frac{1}{r_o}\right)^{-1} = -11.023$$

(d) We perform the same calculations as in parts (a) and and (b) with the new value for W.

$$W := 120 [\![um]\!] :$$
$$R_D := 10 [\![k\Omega]\!] :$$
$$I_D := -\frac{V_{OUT}}{R_D} = -0.000 [\![A]\!] \xrightarrow{\text{replace units}} -250.000 [\![uA]\!]$$
$$V_{OV} := \sqrt{\frac{-I_D}{1.\, \mu p C_{ox} \cdot \frac{W}{L}}} = 0.408 [\![V]\!]$$
$$V_{IN} := V_{DD} - V_{OV} + V_{T0p} = 4.092 [\![V]\!]$$
$$g_m := \frac{2 \cdot (-I_D)}{V_{OV}} = 0.001 [\![S]\!] \xrightarrow{\text{replace units}} 1224.745 [\![uS]\!]$$
$$r_o := \frac{1 + \lambda_p \cdot V_{SD}}{\lambda_p \cdot (-I_D)} = 90000.000 [\![\Omega]\!] \xrightarrow{\text{replace units}} 90.000 [\![kOmega]\!]$$
$$A_v := -g_m \cdot \left(\frac{1}{R_D} + \frac{1}{r_o} \right)^{-1} = -11.023$$

Interestingly, the results from (c) and (d) are the same. This becomes obvious once you realize that the gain is fixed when VOV and VOUT are set (see also problem 2.12). Av = -gmR = 2ID*R/VOV = 2VOUT/VOV. The presence of ro does not change this.

2.15 $W := 10[\![um]\!]:$

(a) $I_D := 100[\![uA]\!] : L := 2[\![um]\!]:$

$$V_{OV} := \sqrt{\frac{I_D}{1.\ \mu nC_{ox} \cdot \frac{W}{L}}} = 0.632\ [\![V]\!]$$

$$g_m := \frac{2 \cdot I_D}{V_{OV}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 316.228\ [\![uS]\!]$$

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.050\ \left[\!\left[\frac{1}{V}\right]\!\right]$$

$$r_o := \frac{1}{\lambda_n \cdot I_D} = 2.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 200.000\ [\![kOmega]\!]$$

$$A_v := g_m \cdot r_o = 63.246$$

(b) $I_D := 50[\![uA]\!] : L := 2[\![um]\!]:$

$$V_{OV} := \sqrt{\frac{I_D}{1.\ \mu nC_{ox} \cdot \frac{W}{L}}} = 0.447\ [\![V]\!]$$

$$g_m := \frac{2 \cdot I_D}{V_{OV}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 223.607\ [\![uS]\!]$$

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.050\ \left[\!\left[\frac{1}{V}\right]\!\right]$$

$$r_o := \frac{1}{\lambda_n \cdot I_D} = 4.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 400.000\ [\![kOmega]\!]$$

$$A_v := g_m \cdot r_o = 89.443$$

(c) $I_D := 100[\![uA]\!] : L := 4[\![um]\!]:$

$$V_{OV} := \sqrt{\frac{I_D}{1.\ \mu nC_{ox} \cdot \frac{W}{L}}} = 0.894\ [\![V]\!]$$

$$g_m := \frac{2 \cdot I_D}{V_{OV}} = 0.000\ [\![S]\!] \xrightarrow{\text{replace units}} 223.607\ [\![uS]\!]$$

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.025\ \left[\!\left[\frac{1}{V}\right]\!\right]$$

$$r_o := \frac{1}{\lambda_n \cdot I_D} = 4.000\ 10^5\ [\![\Omega]\!] \xrightarrow{\text{replace units}} 400.000\ [\![kOmega]\!]$$

$$A_v := g_m \cdot r_o = 89.443$$

We see from this example that there are two ways to increase the intrinsic gain. One is is to decrease the current density ID/W, the other is to increase the channel length.

2.16

$$V_B := 2.5[\![V]\!] : I_B := 500[\![uA]\!] : W := 20[\![um]\!] : L := 1[\![um]\!] : R_D := 5[\![k\Omega]\!] :$$

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.100 \left[\!\left[\frac{1}{V}\right]\!\right]$$

(a) Since VOUT=VB, no current is flowing through the resistor and we can write:

$$I_B = \frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot (V_{IN} - V_{T0n})^2 \cdot (1 + \lambda_n \cdot V_B) :$$

$$V_{IN} := V_{T0n} + \sqrt{\frac{I_B}{\frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot (1 + \lambda_n \cdot V_B)}} = 1.394 [\![V]\!]$$

(b) The value to consider for lambda are:

$$\lambda_{n1} := \lambda_n \cdot (1 + 0.5) = 0.150 \left[\!\left[\frac{1}{V}\right]\!\right]$$

$$\lambda_{n2} := \lambda_n \cdot (1 - 0.5) = 0.050 \left[\!\left[\frac{1}{V}\right]\!\right]$$

The output voltages can be computed using:

$$I_{D0} := \frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot (V_{IN} - V_{T0n})^2 = 0.000 [\![A]\!] \xrightarrow{\text{replace units}} 400.000 [\![uA]\!]$$

$$I_D = I_{D0} \cdot (1 + \lambda_n \cdot V_{OUT}) :$$

$$V_{OUT} = V_B + R_D \cdot (I_B - I_D) :$$

$$V_{OUT} = V_B + R_D \cdot (I_B - I_{D0} \cdot (1 + \lambda_n \cdot V_{OUT})) :$$

$$V_{OUT}(1 + \lambda_n \cdot R_D \cdot I_{D0}) = V_B + R_D \cdot (I_B - I_{D0}) :$$

$$V_{OUT1} := \frac{V_B + R_D \cdot (I_B - I_{D0})}{1 + \lambda_{n1} \cdot R_D \cdot I_{D0}} = 2.308 [\![V]\!]$$

$$V_{OUT2} := \frac{V_B + R_D \cdot (I_B - I_{D0})}{1 + \lambda_{n2} \cdot R_D \cdot I_{D0}} = 2.727 [\![V]\!]$$

As we can see from these results, the quiescent point output voltage varies by about +/-200mV for the given variations in lambda. If this is not acceptable for the intended application of the circuit, we could e.g. opt for a longer channel.

2.17

Note: The problems statement contains errors that have been listed in the errata file. The solution below was generated taking these corrections into account.

$V_{DD} = 5[[V]] : V_{IN} := 1.8[[V]] : W := 15[[um]] : L := 1[[um]] : R_D := 5[[k\Omega]] : R_L := 10[[k\Omega]] :$

(a) We can combine RD and RL into a Thevenin equivalent:

$$V_{th} := 1. \cdot V_{DD} \cdot \frac{R_L}{R_L + R_D} = 3.333 \; [[V]]$$

$$R_{th} := 1. \cdot \frac{R_L \cdot R_D}{R_L + R_D} = 3333.333 \; [[\Omega]]$$

We can now calculate the quiescent point output voltage as follows:

$$I_D := \frac{1}{2} \cdot \mu n C_{ox} \cdot \frac{W}{L} \cdot (V_{IN} - V_{T0n})^2 = 0.001 \; [[A]] \xrightarrow{\text{replace units}} 633.750 \; [[uA]]$$

$$V_{OUT} := V_{th} - R_{th} \cdot I_D = 1.221 \; [[V]]$$

(b)

$$\lambda_n := \frac{\lambda_{Ln}}{L} = 0.100 \left[\left[\frac{1}{V} \right]\right]$$

$$g_m := \mu n C_{ox} \cdot \frac{W}{L} \cdot (V_{IN} - V_{T0n}) \cdot (1 + \lambda_n \cdot V_{OUT}) = 0.001 \; [[S]] \xrightarrow{\text{replace units}} 1094.031 \; [[uS]]$$

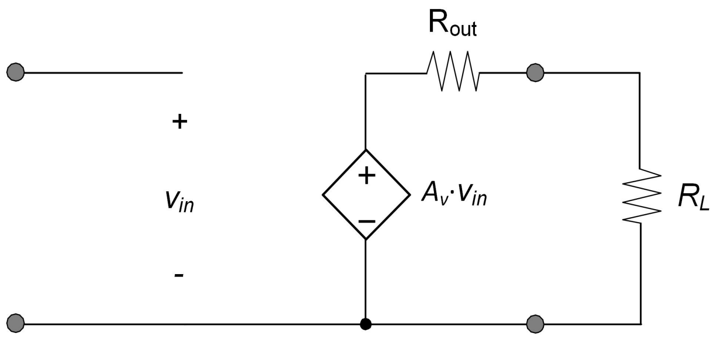
$$r_o := \frac{1 + \lambda_n \cdot V_{OUT}}{\lambda_n \cdot I_D} = 17705.457 \; [[\Omega]] \xrightarrow{\text{replace units}} 17.705 \; [[kOmega]]$$

(c) The open-circuit voltage gain, output resistance and overall voltage gain for the two port model are computed as follows:

$$R_{out} := \frac{r_o \cdot R_D}{r_o + R_D} = 3898.943 \; [[\Omega]]$$

$$A_v := -g_m \cdot R_{out} = -4.266$$

$$A_{vprime} := A_v \cdot \frac{R_L}{R_L + R_D} = -2.844$$



(d) The bias point is now computed without taking RL into account

$V_{OUT} := V_{DD} - R_D \cdot I_D = 1.831 \ [\![V]\!]$

$g_m := \mu n C_{ox} \cdot \frac{W}{L} \cdot (V_{IN} - V_{T0n}) \cdot (1 + \lambda_n \cdot V_{OUT}) = 0.001 \ [\![S]\!] \xrightarrow{\text{replace units}} 1153.547 \ [\![uS]\!]$

$r_o := \frac{1 + \lambda_n \cdot V_{OUT}}{\lambda_n \cdot I_D} = 18668.639 \ [\![\Omega]\!] \xrightarrow{\text{replace units}} 18.669 \ [\![kOmega]\!]$

The two-port model parameters are now:

$R_{out} := \frac{r_o \cdot R_D}{r_o + R_D} = 3943.750 \ [\![\Omega]\!]$

$A_v := -g_m \cdot R_{out} = -4.549$

$A_{vprime} := A_v \cdot \frac{R_L}{R_L + R_D} = -3.033$

We see from this example that connecting RL shifts the output operating point by about 600mV. This causes changes in the small-signal parameters of the MOSFET and hence a shift in the two-pot model parameters. In this example the shift is not dramatic but appreciable. The main conclusion here is that one needs to think about whether the load network will have a significant impact on the amplifier that is to be modeled via a small-signal two-port model. If the answer is yes, then the load network should be connected when computing the bias point, as done in (a)-(b).